Beam

Beam is the basic part of the verniercaliper. Main scale is graduated on beam. Each graduation in main scale equals to 1mm. Every tenth graduation line is marked as bold and big.

Fixed jaw for External or Internal measurement

Fixed jaw is located left side top and bottom of the beam. These two jaws are attached with beam. Vernier unit is moving on the backside of the beam.

Movable jaw for External or Internal measurement

Both movable jaws are moving together during measurement. Main scale is graduated in between the movable and fixed jaw.

Vernier scale moves along with movable jaw in right side. By this movement we can able to measure as required. Vernier unit is locked by locking screw after measuring. To give more accurate measurement a fine adjusting screw is also provided.

Least Count

The least measurement that can be measured with an instrument is called least count. The least count is the difference between main scale division and vernier scale division.

Least count = Main scale division – Vernier scale division

Note

Let	Main scale division	= 1mm
	Vernier scale division	= 9 / 10 = 0.9mm
	Then Least count	= 1-0.9
		= 0.1 mm

Material

Verniercaliper is manufactured with nickel chromium steel.

Depth bar

It is located on the backside bottom of the main scale. It is used to measure the depth of an object.

Procedure for Measurement

- Take the measurement in the main scale
- In vernier scale, note down the line which coincides with main scale.
- Now multiply least count with vernier scale measurement. This gives the vernier scale reading.
- Add main scale reading and vernier scale reading. This is the required measurement.

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